Big Data - MapReduce

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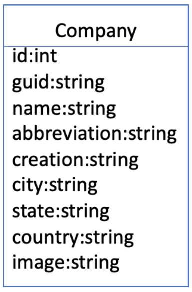
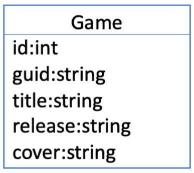
1 Introduction

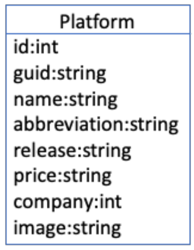
The application to be considered relies on a collection of games, in particular focusing on a set companies, games and platforms. Figures 1a, 1b and 1c present the data schemas.



(a) Data schema Company



(b) Data schema Game



(c) Data schema Platform

Figura 1: Data schemas overview

2 Introducing example

The following example enables to retrieve the number of companies per city.

2.1 Map

```
inKey: numLine
inValue: line (company)

outKey: city
outValue: 1

Mapper(inKey, inValue)
   if (numLine != firstLine)
      outKey=getCity(Line)
      outValue=1
      emitIntermediate(outkey, outValue)
```

2.2 Reduce

```
inKey: city
inValue: list of 1

outKey: city
outValue: number of companies in the city

Reduce(inKey, inValue)
   sum=0;
```

```
foreach element of inValue
    sum++;
outValue=sum
emit(outKey, outValue)
```

3 Activities

1. Aggregation: Number of companies per country.
2. Aggregation: Number of platform per company.
3. Project: Name of the companies.
4. Project: Name of the platforms.
5. Selection: Platform with a price greater than 300$.
6. Aggregation and comparison: Oldest companies per country.
7. Comparison: Newest companies.
8. Aggregation: Cheapest platform.

4 References

- **Lecture**: https://people.irisa.fr/Laurent.Dorazio/data/teachings/big_data/big_data_mapreduce.pdf